College Regulations Assistant: Project Document

1. Problem statement and objective

Students cannot quickly find clear answers about college rules because the information is spread across long handbooks and regulation documents. This project builds a chat assistant that answers student questions about college regulations using only the college's official documents.

Objective: give a student an accurate, short answer in seconds, with a citation to the exact document and page, and a clear refusal when the documents do not cover the question.

Success criteria

- Every answer cites the document name and page number.
- Every answer is traceable to text in the official documents; nothing comes from the model's general knowledge.
- Questions the documents do not cover are declined, with a suggestion of whom to contact.
- The three example questions in the brief (attendance, re-evaluation, hostel visiting hours) are answered correctly.

2. Scope and data sources

The assistant is built on at least two publicly available official documents from your own college or university. Replace the placeholder names below with the real file names.

Document (placeholder name)	Typical content	Example question it answers
Examination Regulations	Attendance eligibility, re-evaluation, grading, deadlines	Minimum attendance to sit the end-semester exam
Student Handbook	Conduct, academic rules, grievance process, contacts	Whom to approach for a grievance

Document (placeholder name)	Typical content	Example question it answers
Hostel Rules	Visiting hours, curfew, leave, discipline	Hostel visiting hours
Library Policy (optional)	Borrowing limits, fines, opening hours	Fine for a late book return

In scope: questions whose answer is stated in these documents.

Out of scope: fees or timetables not in the documents, personal cases (for example, "will I be debarred?"), opinions, and anything needing a staff decision. These are declined with a referral (section 8).

Data rules: use only documents the college has published; keep the original PDFs unchanged; record each file's version or academic year so answers can be checked against it.

3. Solution overview

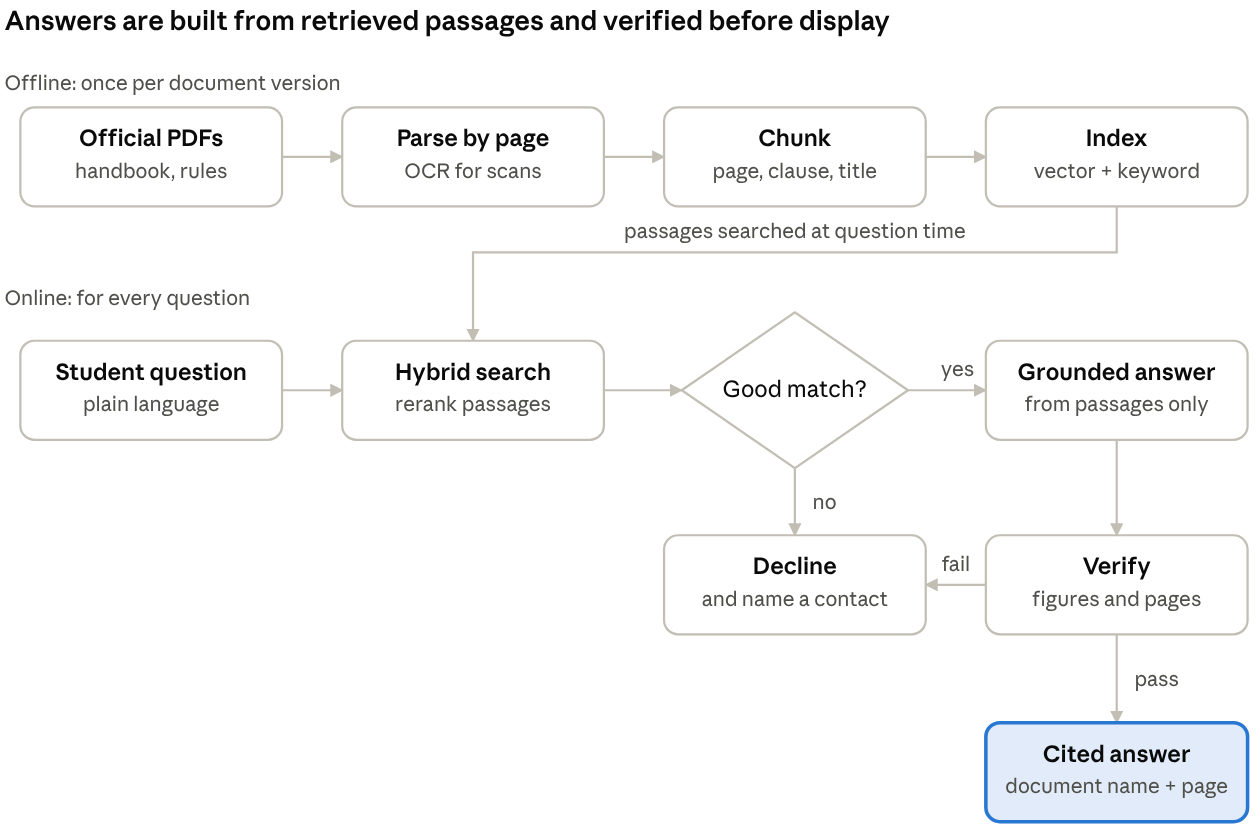
The solution is a retrieval-augmented question-answering assistant: it first finds the relevant passages in the official documents, then writes an answer using only those passages.

Three principles govern every response:

1. **Grounded.** The answer is built only from retrieved passages, never from the model's general knowledge.
2. **Cited.** Every answer ends with the document name and page number of each passage used.
3. **Honest about gaps.** If the passages do not answer the question, the assistant says so and names the office or person to contact.

A student types a question in plain language, for example "Can I apply for re-evaluation and what is the deadline?", and receives a short answer, the supporting citation, and a link to open the source page.

4. System architecture



system architecture · 2 flows, 2 checks

Documents are processed once offline. Each question then passes a confidence check and a verification check; a failure at either point becomes a decline with a contact suggestion.

5. Document ingestion and indexing

Ingestion runs once per document (and again whenever the college publishes a new version). The key rule is that page numbers are kept with every piece of text.

1. **Parse page by page.** Extract text from each PDF page separately so that the page number is known. Scanned PDFs go through OCR first.
2. **Clean.** Remove repeated headers, footers and page numbers; keep headings, clause numbers and tables.
3. **Chunk.** Split text into passages of about 200 to 400 words along clause or heading boundaries, with a small overlap. A passage never crosses a page boundary unless a clause continues; in that case both pages are recorded.

4. **Attach metadata.** Each chunk stores: document name, document version or year, page number(s), section heading and clause number.
5. **Index.** Create a vector embedding for each chunk (for meaning-based search) and a keyword index (for exact terms such as "75%" or "re-evaluation"). Store both in a vector database.

Example stored chunk:

Field	Value
document	Examination Regulations 2025-26
page	14
section	7.2 Eligibility to appear
text	(clause text exactly as printed)

6. Retrieval and answer generation

Each question follows the same path, which keeps answers consistent and checkable.

1. **Understand the question.** Rewrite it as a clear search query (for example, expand "attendance rule for exams" to include "minimum attendance", "eligibility", "end-semester examination").
2. **Hybrid retrieval.** Run vector search and keyword search together, then merge the results. This catches both paraphrased and exact-term matches.
3. **Rerank.** A reranking model scores the top 10 to 20 passages and keeps the best 3 to 5.
4. **Confidence check.** If the best passage scores below a set threshold, skip generation and go straight to the decline path (section 8).
5. **Grounded generation.** The language model receives only the selected passages and a strict instruction: answer from these passages only, quote or paraphrase closely, and cite the passage you used. If the passages do not contain the answer, reply "not covered".
6. **Verify.** A final check confirms that each cited page appears in the retrieved passages and that key facts in the answer (numbers, dates, days) appear in the cited text. A failed check triggers a decline instead of a guess.

Answer format

- One or two sentences that answer the question directly.
- Any condition or exception stated in the document.
- Source line: document name, page number, and section.

7. Citation mechanism

Citations come from stored metadata, not from the language model's memory, so they cannot be invented.

- Each retrieved chunk carries its document name and page number from ingestion.
- The model is told to refer to passages by number (for example [1], [2]); the application then replaces each number with the stored document name and page.
- A source panel shows the exact passage text beside the answer, with a link that opens the PDF at that page.
- If an answer uses passages from two documents or pages, all are listed.
- An answer with no valid citation is never shown; it becomes a decline.

Citation format shown to the student:

```
Source: Examination Regulations 2025-26, page 14, Section 7.2
```

8. Out-of-scope handling and referral

The assistant declines when retrieval confidence is low, the answer check fails, or the question needs a personal or discretionary decision. It never guesses.

A decline has three parts: a plain statement that the documents do not cover it, what the documents do say that is nearby (if anything), and whom to contact.

Topic of the question	Contact to suggest
Exams, attendance shortage, re-evaluation	Controller of Examinations / Examination Cell
Hostel allotment, rules, complaints	Hostel Warden
Library fines, borrowing	Librarian
Fees and scholarships	Accounts / Fee Section
Academic or course issues	Head of Department or Class Advisor

Topic of the question	Contact to suggest
Anything else	Student Affairs / Dean of Students

The contact list is a configuration file filled in with your college's real offices, email addresses and phone numbers from the official handbook.

Example decline

> I could not find this in the college regulations I have access to, so I cannot answer it reliably. Please contact the Examination Cell for this.

9. Technology stack and user interface

Layer	Suggested choice	Why
PDF parsing	PyMuPDF or pdfplumber (OCR: Tesseract for scans)	Page-level text extraction
Chunking and orchestration	Python with LangChain or LlamaIndex	Ready-made RAG components
Embeddings	Multilingual sentence-embedding model	Handles English and regional-language questions
Vector store	Chroma or FAISS, plus BM25 keyword index	Free, runs locally
Reranker	Cross-encoder reranking model	Better top results
Answer model	Claude through the API, with a strict grounded prompt	Follows "answer only from the passages" instructions well
Backend	FastAPI	Simple API for the web page
Frontend	React or Streamlit web page	Chat box, source panel, PDF page viewer

Interface

- A chat box with a few suggested questions.
- The answer, then a "Source" line with document name and page.

- An expandable panel showing the exact passage, and a button to open the PDF at that page.
- A "not covered" card with the contact details when the assistant declines.
- A footer note: "Answers come from official documents; confirm important decisions with the office."
- An admin page to upload a new document version and re-run indexing.

10. Sample questions and expected answers

The figures below are illustrative only. The live assistant must return the values printed in your college's own documents.

Student question	Expected behaviour	Example response format
What is the minimum attendance required to sit the end-semester examination?	Answer with the percentage and any condonation rule	"A student needs at least [X]% attendance in each course. [Condonation rule, if any.] Source: Examination Regulations, page [N]"
Can I apply for a re-evaluation, and what is the deadline?	Answer yes or no, the procedure, and the deadline	"Yes, [eligibility]. Apply within [N] days of result publication. Source: Examination Regulations, page [N]"
What are the hostel visiting hours?	Give the hours and any exceptions	"Visitors are allowed from [time] to [time]. Source: Hostel Rules, page [N]"
Will I be allowed to sit the exam if my attendance is 70%?	Decline the personal case; state the general rule and refer	"I cannot decide individual cases. The general rule is [rule]. Source: [..]. Please contact the Examination Cell."
What is the canteen menu today?	Decline: not in the documents	"This is not covered in the college regulations. Please contact [Student Affairs]."

11. Evaluation plan

Accuracy is checked against a test set written by hand from the documents before the demo.

Test set (about 60 questions)

- 35 answerable questions across all documents, each with the correct answer and the correct page noted.
- 10 paraphrased versions of the same questions (different wording, short forms, spelling mistakes).
- 15 unanswerable questions (not in the documents, personal cases, or unrelated topics).

Metric	How it is measured	Target
Answer correctness	Answer matches the document on a manual check	90% or higher
Citation accuracy	Cited document and page contain the answer	95% or higher
Correct decline rate	Unanswerable questions are declined	95% or higher
False decline rate	Answerable questions wrongly declined	10% or lower
Response time	Question to answer	Under 5 seconds

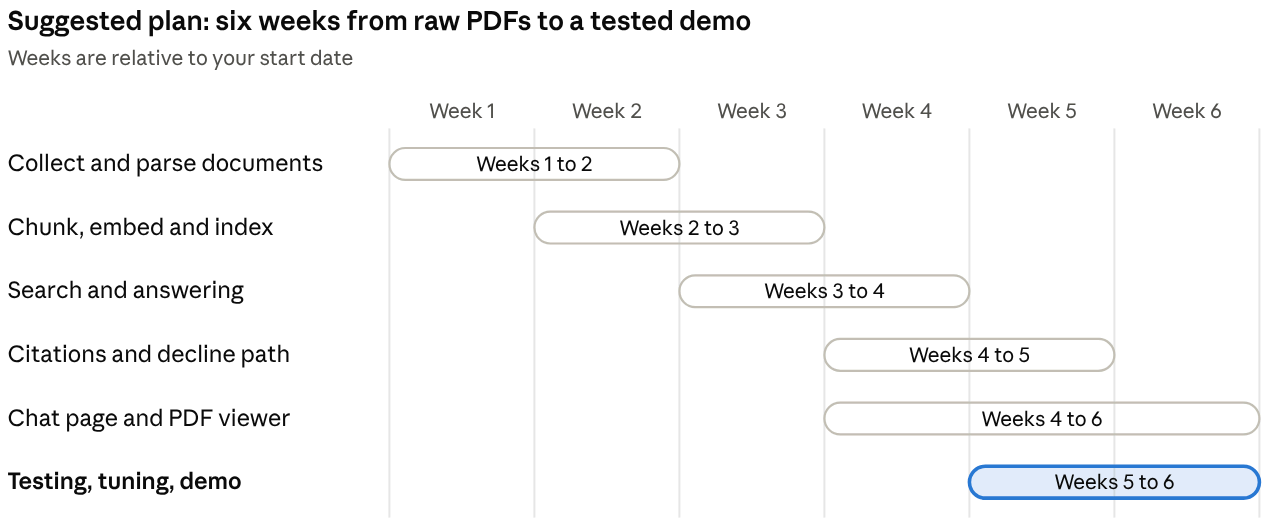
Failures are reviewed, then the chunking, thresholds or prompt are adjusted and the test set is run again.

12. Risks and mitigations

Risk	Mitigation
The model invents a rule or number	Answer only from retrieved passages; verify numbers and dates against the cited text; decline on failure
Wrong or missing page number	Page numbers come from stored metadata, not from the model

Risk	Mitigation
Scanned or badly formatted PDFs lose text	OCR, then spot-check extracted text page by page
Rules in a table are split across chunks	Keep tables whole as single chunks
Documents change each year	Store the version with each chunk; re-index when a new version is published
Students treat answers as final decisions	Footer note and decline of personal cases; point to the responsible office
Conflicting rules in two documents	Show both with their citations and say they differ

13. Implementation timeline and deliverables



implementation plan · 6 workstreams over 6 weeks

The plan assumes one small team. Interface work starts in week 4 so it can overlap with answer tuning. Shift the weeks to fit your own deadline.

Deliverables checklist

- [ ] Working web assistant: chat page, source panel, PDF viewer
- [ ] At least two official college documents, parsed and indexed
- [ ] Source code and setup instructions
- [ ] Test set with results against the targets in section 11

- [ ] Demo covering the three example questions and at least two declined questions
- [ ] This project document